

Wet Lab Outsourcing — Execution Plan

Microplastic Degradation Research Project | Researcher: Arjun Popuri | April 2026

This document is the step-by-step execution plan for validating the engineered IsPETase variant without requiring direct lab access. Each step identifies what to do, where to go, what to send, how much it costs, and what evidence it produces for the research paper and PI pitch. Execute in order of priority.

Master execution plan — all 6 steps at a glance

Priorit y	Step	Action	Cost	Timeline	Lab needed?
1	Gene synthesis	Order from Twist Bioscience	~\$162	2 weeks	NO
2	Protein expression	Email iGEM teams UTD/UT Austin	Free	4-6 weeks	Collaboration
3	LC-MS/MS proteomics	Rapid Novor commercial service	\$300-800	2 weeks	NO
4	nanoDSF Tm	UT Southwestern core facility	\$100-300	1 week	Core only
5	SGF incubation + HPLC	DIY incubation + HPLC service	\$200-500	Parallel	DIY possible
6	PET/BHET activity assay	DIY with BHET substrate	\$50-100	Parallel	DIY possible

Step 1 — Gene Synthesis

NOW

No lab access required. You have two 903 bp FASTA files ready to order today. This is the single most important action to take first — having synthesized genes in hand transforms your PI pitch from theoretical to operational.

What you are ordering

Two codon-optimized expression constructs, each 903 bp, ready for cloning into a pET expression vector. Both are in your results/step5_constructs/ folder.

Files to submit:

- [] results/step5_constructs/engineered_IsPETase_cloning_construct.fasta (903 bp)
- [] results/step5_constructs/wildtype_IsPETase_cloning_construct.fasta (903 bp)

Vendors

Vendor	URL	Cost	Notes
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Twist Bioscience	twistbioscience.com	~\$81 each	Fastest turnaround, highly reliable
IDT	idtdna.com/pages/products/gene-and-gene-fragments/gene-synthesis	~\$90 each	Good for smaller quantities
Genscript	genscript.com/gene-synthesis.html	~\$95 each	Includes sequence verification

Exact steps to order from Twist Bioscience

- [] Create a free account at twistbioscience.com
- [] Click Products > Gene Fragments > Gene Synthesis
- [] Upload engineered_IsPETase_cloning_construct.fasta
- [] Select: cloning vector = pET-28a(+), host = E. coli
- [] Add to cart, repeat for wildtype_IsPETase_cloning_construct.fasta
- [] At checkout: specify E. coli SHuffle T7 as intended expression host in notes
- [] Submit order — receive confirmation email with tracking
- [] Receive both genes in ~10-14 business days as glycerol stabs or DNA in tubes

What evidence this produces for your paper: Gene synthesis confirmation receipt proves the constructs were validated as synthesizable — commercial gene synthesis companies run sequence verification on every construct before shipping. If Twist accepts your sequence without flagging errors, that is independent third-party validation of your construct design.

Step 2 — Protein Expression via iGEM Collaboration

NOW

iGEM (International Genetically Engineered Machine) teams at universities have full *E. coli* expression infrastructure and actively seek interesting projects. This is the fastest free path to expressed and purified protein.

Target iGEM teams to contact

Team	URL	Why contact them	Contact method
UT Dallas iGEM	igem.org/team/UTDallas	Closest to The Colony, protein engineering focus	igem.utdallas.edu
UT Austin iGEM	igem.org/team/UT-Austin	FAST-PETase creators nearby, PETase expertise	utigem.com
Rice University iGEM	igem.org/team/Rice	Strong synthetic biology program	igem.rice.edu

What to ask them for

- [] Transform your synthesized pET plasmid into *E. coli* SHuffle T7
- [] Grow overnight culture, induce with 0.5 mM IPTG at 18 degrees C for 16 hours
- [] Harvest cells by centrifugation at 4000 rpm for 20 minutes
- [] Lyse cells by sonication (30 seconds on, 30 off, 10 cycles)
- [] Purify by nickel affinity chromatography using the N-terminal 6xHis tag
- [] Elute with 250 mM imidazole, dialyze into PBS pH 7.4
- [] Measure concentration by BCA assay or A280
- [] Run SDS-PAGE to confirm purity (expected band at ~33 kDa for IsPETase + tag)
- [] Ship purified protein on dry ice to address you specify

Alternative: commercial expression service

Vendor	URL	Cost	Turnaround
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GenScript	genscript.com/protein-expression-service.html	\$500-1500	3-6 weeks
Creative Biostructure	creative-biostructure.com	\$800-2000	4-8 weeks
Twist Bioscience	twistbioscience.com/protein	\$600-1200	4-6 weeks

Important: specify E. coli SHuffle T7 as the expression strain in all communications. Standard BL21 or DH5-alpha strains will break your T51C/T72C disulfide bond. SHuffle T7 is the only strain that allows cytoplasmic disulfide formation.

Step 3 — LC-MS/MS Proteomics — Pepsin Cleavage Validation

AFTER PROTEIN

This experiment directly validates your L117I pepsin resistance mutation by showing the L117 cleavage fragment is present in the wild-type SGF digest but absent in the engineered variant digest. It is fully outsourceable to a commercial mass spectrometry service — no lab bench required beyond preparing the samples.

What you need before doing this step

- ☐ Purified wild-type IsPETase (from Step 2)
- ☐ Purified engineered IsPETase variant (from Step 2)
- ☐ Pepsin (purchasable from Sigma-Aldrich, catalog P7012)
- ☐ HCl to make pH 1.2 buffer
- ☐ Dry ice for shipping samples

How to prepare samples yourself (no lab bench needed)

- ☐ Make simulated gastric fluid: 0.2% NaCl + 0.32% pepsin, adjust to pH 1.2 with HCl
- ☐ Mix 50 ug of each protein with 500 uL of SGF in separate microtubes
- ☐ Incubate at 37 degrees C for 60 minutes in a water bath
- ☐ Stop reaction by adding 50 uL of 1M NaHCO₃ (raises pH, inactivates pepsin)
- ☐ Freeze immediately at -80 degrees C or on dry ice
- ☐ Ship frozen on dry ice to your chosen vendor

Commercial LC-MS/MS vendors

Vendor	URL	Cost per sample	Notes
Rapid Novor	rapidnovor.com	\$150-400	Fast turnaround, good peptide coverage

Creative Proteomics	creative-proteomics.com	\$200-500	Full peptide mapping service
Alphalyse	alphalyse.com	\$300-600	Specializes in protein ID and mapping
UT Southwestern Core	utsouthwestern.edu/research/core-facilities	\$80-200	Cheapest if you can access

What to ask the vendor for

- [] Bottom-up proteomics with trypsin digest of the submitted samples
- [] LC-MS/MS peptide identification with >95% confidence threshold
- [] Peptide coverage map showing all identified peptides
- [] Specifically flag presence or absence of the peptide spanning position 117
- [] Deliver results as PDF report + raw data file

What evidence this produces: If the peptide containing L117 appears in the wild-type digest but not the engineered variant digest, this is direct molecular proof that L117I eliminated pepsin cleavage at that site. This is publication-quality evidence for your pepsin resistance mutation.

Step 4 — nanoDSF Thermal Stability Measurement

AFTER PROTEIN

nanoDSF measures the melting temperature (T_m) of both variants by tracking tryptophan fluorescence as temperature increases. Higher T_m in the engineered variant confirms the T51C/T72C disulfide bond improves thermal stability. Available through core facilities without a full lab relationship.

What you need before this step

- [] 20-50 uL of purified protein per variant at 0.5-1.0 mg/mL concentration
- [] Protein in a clean buffer (PBS pH 7.4 or 50mM sodium phosphate pH 7.4)
- [] Ship on dry ice if mailing, or bring on ice if visiting in person

Where to access nanoDSF

Facility	URL	Cost	Notes
UT Southwestern Biophysics Core	utsouthwestern.edu/research/core-facilities	\$50-200	External users welcome, fee schedule online
NanoTemper Prometheus Service	nanotempertech.com/prometheus	\$200-500	Mail-in service, no affiliation needed
UT Dallas Shared Equipment	utdallas.edu/research/core-research	Contact for rates	May require faculty sponsor
Rice University Core	research.rice.edu/cores	\$100-300	External access available

Exact measurement protocol to request

- [] Instrument: NanoTemper Prometheus NT.48 or equivalent nanoDSF system
- [] Temperature ramp: 20 to 95 degrees C at 1 degree per minute
- [] Measurement: F350/F330 fluorescence ratio (tryptophan unfolding signal)
- [] Run both engineered and wild-type in the same experiment for direct comparison
- [] Request: T_m value, onset temperature, and full melting curve data export
- [] Also request: run at pH 7.4 (standard) and pH 5.0 (partial acid stress) if possible

What evidence this produces: A higher T_m for the engineered variant (even 2-5 degrees C) confirms the disulfide bond and charge-reversal mutations improve structural stability. This directly validates Predictions 1 and 2 from your computational study and is the single most important number your paper needs.

Step 5 — SGF Incubation + HPLC TPA Quantification

PARALLEL

This is the central experiment that directly tests your hypothesis. The SGF incubation can be done without a research lab. The HPLC analysis is outsourced to a commercial analytical chemistry service.

What to buy (all available on Sigma-Aldrich or Amazon)

- [] Pepsin from porcine stomach mucosa (Sigma-Aldrich P7012, 250mg, ~\$35)
- [] Hydrochloric acid 1M solution (Sigma-Aldrich, ~\$25)
- [] Sodium chloride (table salt or lab grade, ~\$10)
- [] Sodium bicarbonate (baking soda works, ~\$5)
- [] 1.5 mL microtubes (Amazon, ~\$10 for 500)
- [] Digital thermometer accurate to 0.1 degree C (~\$15)
- [] 37 degree C water bath or incubator (Amazon, ~\$80-150)
- [] pH meter (Amazon, ~\$20-30 for basic model)

Simulated gastric fluid recipe

- [] Dissolve 2.0 g NaCl in 900 mL distilled water
- [] Add 3.2 g pepsin powder and stir until dissolved
- [] Adjust pH to exactly 1.2 using 1M HCl solution (use pH meter)
- [] Bring total volume to 1000 mL with distilled water
- [] Use immediately or store at 4 degrees C for up to 24 hours

SGF incubation protocol

- [] Label 10 microtubes: WT-0min, WT-15min, WT-30min, WT-60min, WT-ctrl,
- [] ENG-0min, ENG-15min, ENG-30min, ENG-60min, ENG-ctrl
- [] Add 5 uL purified protein (1 mg/mL) to each tube
- [] Add 45 uL SGF to each tube except the two -ctrl tubes (add PBS instead)
- [] Incubate all tubes at 37 degrees C in water bath
- [] At each time point (0, 15, 30, 60 min): remove corresponding tubes
- [] Add 5 uL 1M NaHCO₃ to stop the pepsin reaction immediately
- [] Flash freeze on dry ice and store at -80 degrees C until HPLC analysis

HPLC TPA analysis vendors

Vendor	URL	Cost	Notes
Intertek	intertek.com/pharmaceutical/analytical	\$100-300 per run	Pharmaceutical analytical lab
Eurofins	eurofins.com/food-testing	\$80-250 per run	Large network, fast turnaround
SGS	sgs.com/en/laboratories	\$100-300 per run	ISO certified results
Chemir	chemir.com	\$150-400 per run	Specializes in polymer analysis

What to ask the HPLC vendor

- [] Reverse-phase HPLC with UV detection at 260 nm
- [] Mobile phase: 30% acetonitrile in 0.1% formic acid (standard for TPA detection)
- [] Quantify terephthalic acid (TPA) in each sample against a TPA standard curve
- [] TPA standard: available from Sigma-Aldrich catalog 185361
- [] Report results as TPA concentration in μM or nmol/mL for each time point
- [] Ship samples frozen on dry ice with sample list

What evidence this produces: A plot of residual PET-hydrolyzing activity over time for both variants. If the engineered variant retains greater than 50% activity after 60 minutes while wild-type loses most activity, your central hypothesis is confirmed. This is the most important result in your entire study.

Step 6 — PET/BHET Activity Assay — Retained Catalytic Function

PARALLEL

This confirms your engineered variant still degrades PET at neutral pH — proving the mutations did not disrupt the active site. Use BHET as substrate instead of PET film for simplicity. BHET is commercially available and dissolves in buffer unlike solid PET film.

What to buy

- [] BHET (bis-2-hydroxyethyl terephthalate) from Sigma-Aldrich catalog 465151, ~\$35
- [] 50 mM sodium phosphate buffer pH 7.0 (make from sodium phosphate + HCl)
- [] DMSO to dissolve BHET (Sigma-Aldrich, ~\$20)

BHET activity assay protocol

- [] Dissolve BHET in DMSO at 10 mM concentration (stock solution)
- [] Dilute to 1 mM BHET in 50 mM sodium phosphate pH 7.0 (max 1% DMSO final)
- [] Add 5 uL purified protein (1 mg/mL) to 495 uL BHET solution in microtube
- [] Incubate at 30 degrees C for 0, 1, 2, 4, 8, and 24 hours
- [] At each time point remove 50 uL and add to 50 uL methanol to stop reaction
- [] Freeze immediately, ship to HPLC vendor for TPA quantification
- [] Include a no-enzyme control (buffer only + BHET) as background
- [] Include a boiled enzyme control (heat at 95 C for 10 min before adding) as negative control

What evidence this produces: If both variants produce TPA from BHET at pH 7.0 at comparable rates, catalytic activity is confirmed to be retained. If the engineered variant shows similar or improved TPA production versus wild-type at neutral pH, this directly validates Prediction 4 from your computational study and the R280E subsite IIc finding.

Step PI Pitch — How This Changes Your Lab Outreach Email

NOW

Before outsourcing (what you have now)

I have a computationally designed 8-mutation IsPETase variant with cloning-ready constructs and am looking for a lab to validate it experimentally.

After completing Steps 1-3 (what you could have in 6-8 weeks)

I have a computationally designed 8-mutation IsPETase variant with synthesized genes in hand. Protein expression is underway through a collaboration with the UT Dallas iGEM team. I have arranged LC-MS/MS proteomics through Rapid Novor to validate the L117I pepsin resistance mutation. I am seeking access to nanoDSF and HPLC instrumentation to complete the thermal stability and activity characterization. All other aspects of the validation plan are in progress independently.

That second email gets a response. The professor sees a student who does not need hand-holding — just instrument access. That is the easiest possible yes for a PI.

Master to-do checklist — print and track progress

- [] ORDER: Submit both FASTA files to Twist Bioscience for gene synthesis (\$162)
- [] EMAIL: Contact UT Dallas iGEM team with expression request and protocol
- [] EMAIL: Contact UT Austin iGEM team as backup expression option
- [] BUY: Pepsin, HCl, NaCl, NaHCO₃, microtubes, pH meter, 37C water bath
- [] RECEIVE: Synthesized genes from Twist (~2 weeks)
- [] RECEIVE: Purified protein from iGEM collaboration (~4-6 weeks)
- [] PREPARE: SGF incubation samples using DIY protocol above
- [] SHIP: SGF samples to Rapid Novor or Creative Proteomics for LC-MS/MS
- [] CONTACT: UT Southwestern Biophysics Core for nanoDSF appointment
- [] PREPARE: BHET activity assay samples at pH 7.0
- [] SHIP: Activity samples to HPLC analytical service
- [] COLLECT: All results and compile into results section of paper
- [] UPDATE: PI outreach emails with new experimental progress